ABDUL WALI KHAN UNIVERSITY MARDAN

Department of Computer Science

FYP MANAGEMENT SYSTEM

Complete Project Documentation

SUBMITTED BY

MS

Muhammad Sudais

Full-Stack Developer

AI

Atiya Inayat

Developer

MA

Mehboob Ali

Frontend Developer

VERSION	DATE	THEME
1.0	April 2026	AWKUM Official

Table of Contents

1.	Project Overview	3
2.	Objectives & Scope	4
3.	User Roles	5
4.	Features by Role	6
4.1	Public Pages	6
4.2	Student Dashboard	7
4.3	Supervisor Dashboard	8
4.4	Admin Dashboard	9
5.	System Workflow	10
6.	Marks Evaluation Criteria	11
7.	Technical Architecture	12
8.	Project Structure	13
9.	Theme System	14
10.	Special Features	15
11.	ID Generation System	16
12.	How to Use the System	17
13.	Future Enhancements	18
14.	Conclusion	19

1. Project Overview

What this system is and why it exists

The **FYP Management System** is a centralized academic platform designed for the Department of Computer Science at **Abdul Wali Khan University Mardan (AWKUM)**. It streamlines the entire Final Year Project lifecycle — from initial proposal submission and team formation, to supervisor allocation, mid-term evaluations, and final result announcements.

Traditionally, FYP coordination involves scattered emails, paper forms, manual marks calculation, and frequent miscommunication between students, supervisors, and the department. This system replaces all of that with a unified, role-based digital workspace where every stakeholder has clear visibility into their responsibilities and progress.

Key Goals

- **Eliminate paperwork** — All proposals, reports, and marks are submitted and tracked digitally.
- **Provide transparency** — Students can see their marks, deadlines, and status at any time.
- **Centralize communication** — Replace scattered emails with structured in-app messaging.
- **Automate ID generation** — Auto-create unique Student, Team, and Supervisor IDs.
- **Standardize evaluation** — Enforce a consistent 200-mark evaluation rubric across all projects.
- **Empower administrators** — Give the department head full oversight without micromanagement.

Why This Matters

FYP is one of the most defining academic experiences in a CS degree. A well-organized system ensures students focus on **building their projects** rather than chasing paperwork — and gives faculty the tools they need to fairly evaluate dozens of teams every semester.

2. Objectives & Scope

What the system does and doesn't do

Primary Objectives

1. Provide a **single point of access** for all FYP-related activities at AWKUM CS Department.
2. Enable students to **register teams**, submit proposals, and track their project progress.
3. Allow supervisors to **review submissions**, assign marks, and monitor multiple teams simultaneously.
4. Give administrators full **oversight and control** over supervisor allocations and academic records.
5. Implement a **structured 200-mark evaluation system** covering meetings, submissions, and viva.
6. Maintain a **permanent digital record** of every team, project, and academic outcome.

In Scope

- Public-facing pages: Home, Coursera Requirements, FAQs, Authority Messages, Developer Profiles
- Three role-based portals: Student, Supervisor, Admin
- Team formation flow with auto-generated IDs (max 3 students per team)
- Proposal upload with state tracking (Pending, Approved, Rejected)
- File upload system supporting reports, code, and documentation
- Marks management with live validation and per-component breakdowns
- Search functionality for teams and students (by ID or name)
- Picture-in-picture promotional video with controls
- Themeable design system supporting multiple color palettes

Out of Scope (For This Phase)

- Live video conferencing (use Zoom/Teams externally)
- Plagiarism checking (handle externally via Turnitin)
- Public project showcase / past-FYP archive
- Mobile native apps (web is fully responsive instead)
- Multi-department support (CS-only for v1.0)

3. User Roles

Three actors, three views

The system is built around **three distinct user roles**. Each gets their own dashboard, permissions, and workflow tailored to what they actually need to do.

■■■ STUDENT

Final-year CS undergraduate forming a team and working on a project

Can do:

- Create a team and invite up to 2 additional members
- Choose a supervisor from the approved list
- Submit and re-upload project proposal
- Upload reports, code files, and documentation
- View own marks and team performance
- Track deadlines and academic schedule

■■■ SUPERVISOR

Faculty member assigned to guide one or more student teams

Can do:

- View all assigned teams and their members
- Search teams by Team ID or project title
- Search students by Student ID or name
- Approve or reject team proposals
- Assign marks across 11 evaluation components (per student)
- Provide feedback through structured comments

■■ ADMIN

Department head or FYP coordinator with full system oversight

Can do:

- Add new supervisors with auto-generated credentials
- Reassign teams to different supervisors
- Click any supervisor card to view all their teams & students
- Edit any student's marks (full read-write access)
- Browse all students and teams across the department
- Monitor system-wide statistics and submission rates

4. Features by Role

Detailed page-by-page breakdown

4.1 Public Pages (No Login Required)

These pages are accessible to anyone visiting the site — prospective students, parents, faculty, or visitors. They provide general information about the FYP process and the department.

Feature	Description
Home Page	Hero banner with image slider, system overview cards, statistics row (students/supervisors/projec
Coursera Requirements	Dedicated page listing all approved Coursera courses by category. Students must complete 8 cer
FAQs	Expandable accordion-style answers organized into three categories: Coursera Courses, Starting
Messages	Official communications from authorities — the Department Chairman and the FYP Coordi
Developers	Profiles of the three developers behind the system: Muhammad Sudais (Full-Stack), Atiya Inayat
PiP Video	A small floating video plays in the bottom-right corner on every public page. Users can mute/unm

4.2 Student Dashboard

After logging in as a student, users land on a five-tab dashboard tailored to their FYP journey.

Feature	Description
Home	Welcome banner with auto-generated Student ID and Team ID badges, plus a re
My Team	Full team details including Team ID, project title, creation date, status, and a list of all team memb
My Project	Project details, proposal submission area (with state machine: idle → pending → approved/reject
Supervisor	Read-only profile of the assigned supervisor: name, email, office location, office hours, and resea
Progress & Evaluation	Per-student tabs to view marks. Shows total score (out of 200), percentage ring, animated progre
3-Step Setup	On first login, a guided modal opens: Step 1 — enter project title & category;

4.3 Supervisor Dashboard

Supervisors get a focused two-tab interface optimized for managing multiple teams efficiently.

Feature	Description
Welcome Block	Personalized greeting with the supervisor's auto-generated SUP ID displayed in a gold-b
Statistics Strip	Three live stat cards: number of assigned Groups, total Students across all groups, and the Total
My Groups Card	Tabular list of all assigned teams with live search by Team ID or project title. Search mat
All Students Card	Combined list of every student under the supervisor's care. Includes live search by Student ID
Group Detail View	Per-group page showing project info, supervisor reference, and an expandable card for each stud
Marks Validation	Real-time validation: empty inputs show as 'pending' (dashed border), invalid values (over the ma
Empty Search State	When a search returns no results, a friendly empty state appears (with magnifying-glass icon) sho

4.4 Admin Dashboard

The admin dashboard gives the FYP coordinator complete control over the system. Built around five views accessible from a collapsible sidebar.

Feature	Description
Dashboard Overview	Welcome banner with quick action button, three high-level stat cards (Total Students, Supervisors
Add Supervisor Modal	Form to register a new supervisor with: auto-generated SUP ID (cannot be edited), full name, inst
Supervisors View	Grid of all faculty cards. Each card shows: name, email, SUP ID, team count, available capacity.
Supervisor Detail View	Header with supervisor avatar & stats. Below: every team assigned to that supervisor render
Teams View	Tabular list of all teams in the department. Each row has a Reassign Supervisor action (c
Students View	All registered students with their team, supervisor, current marks, and edit access.
Marks & Evaluation View	System-wide marks overview with summary stats (Avg. Score, Top Score, Pending Reviews, Cor

5. System Workflow

Step-by-step user journey

Typical FYP Lifecycle in This System

1

Admin creates supervisor accounts

Department head adds faculty members through the Add Supervisor modal. Each gets an auto-generated SUP ID (e.g. SUP-20260001) and a temporary password emailed to them.

2

Student logs in for the first time

Student authenticates and is greeted by the 3-step setup wizard. They see their auto-generated Student ID with a save-this-ID reminder.

3

Student creates a team

Step 1: enter project title & category. System generates a unique Team ID like TEAM-20261284 with a prominent notice asking them to save it.

4

Student adds members

Step 2: search team members by their Student IDs. Up to 2 additional members can be added (max team size = 3, including leader).

5

Student picks a supervisor

Step 3: select from a dropdown showing each supervisor's research areas. The system shows a preview card before confirmation.

6

Student uploads proposal

From the My Project tab, student uploads their proposal PDF. State changes from **idle** → **pending** automatically.

7

Supervisor reviews proposal

Supervisor logs in, opens the team's detail page, and either approves or rejects. Status updates instantly visible to the student.

8

Monthly meetings & submissions

Throughout the year, supervisor records attendance/quality marks for monthly meetings (Oct, Nov, Dec, Feb, Mar) and grades submissions (proposal: 10, progress report: 20).

9

Mid & final viva evaluation

Internal Viva 1 (25 marks) at mid-evaluation. Internal Viva 2 (25 marks) and External Viva (50 marks) at the end.

10 **Final documentation**
Student uploads final documentation. Supervisor grades it (20 marks). Total max = 200.

11 **Results announcement**
Final grades visible in the student's Progress & Evaluation tab. Admin can view, edit, or audit any record.

6. Marks Evaluation Criteria

How the 200 total marks are distributed

Every FYP student is evaluated on a **standardized 200-mark rubric** divided across four sections and 11 specific components. This ensures fair, consistent grading across all supervisors and projects.

Section	Component	Max Marks
Monthly Meetings	October	10
	November	10
	December	10
	February	10
	March	10
Submissions	Proposal Submission	10
	Progress Report	20
Viva	Internal Viva 1	25
	Internal Viva 2	25
	External Viva	50
Documentation	Final Documentation	20
	GRAND TOTAL	**200**

Key Insight

Viva contributes 100 marks (50%) — making it the single biggest factor. Students should never neglect their final defense preparation. Monthly meetings, while individually small, add up to **50 marks (25%)** — consistent attendance throughout the year is essential. The Final Documentation accounts for **20 marks (10%)**.

7. Technical Architecture

How the system is built

Technology Stack

Frontend	HTML5, CSS3, Vanilla JavaScript (no framework dependencies)
Charts	Chart.js 4.4.2 — used for marks visualization
Fonts	Calibri (with Segoe UI / Trebuchet MS / Arial fallbacks)
Icons	Inline SVG (zero external dependencies, fully customizable)
Theme System	CSS Custom Properties (CSS variables) with shared theme.css
State Storage	JavaScript objects in memory (browser localStorage for sessions)
Authentication	Role-based login redirecting to dedicated dashboards
Responsive Design	Mobile-first with breakpoints at 600px, 768px, 1024px

Design Principles

- **No build step required** — Just open index.html in any browser. Easy for evaluators to test.
- **Theme-driven design** — Every color and shadow uses CSS variables. Switch the entire site by editing one file.
- **Component reusability** — Cards, buttons, badges, and tables all share consistent styling classes.
- **Accessibility first** — Proper labels, ARIA attributes, keyboard navigation, semantic HTML.
- **Progressive disclosure** — Don't show everything at once. Use accordions, modals, and tabs to reveal info as needed.
- **Mobile parity** — Hamburger menus, stacked layouts, touch-friendly buttons throughout.

8. Project Structure

How the codebase is organized

```
FYP-Management-System/
├── Client/
│   ├── index.html                ← Public landing page
│   ├── images/                   ← University photos, slider, developers
│   ├── shared/
│   │   ├── css/
│   │   │   ├── theme.css         ← THE master theme file (edit to change colors)
│   │   │   └── global.css        ← Public site styles
│   │   ├── js/
│   │   │   └── script.js         ← Public site logic
│   │   └── videos/
│   │       └── intro.mp4         ← PiP video file
│   ├── Auth/
│   │   ├── login.html
│   │   ├── css/auth.css
│   │   └── js/auth.js
│   ├── Student Dashboard/
│   │   ├── std.html
│   │   ├── css/student.css
│   │   └── js/student.js
│   ├── Supervisor Dashboard/
│   │   ├── supervisor.html
│   │   ├── css/supervisor.css
│   │   └── js/supervisor.js
│   └── Admin Dashboard/
│       ├── admin.html
│       ├── css/admin.css
│       └── js/admin.js
└── Documentation/
    ├── FYP_Theme_Palette.pdf
    └── FYP_Management_System_Documentation.pdf
```

Each role has its own self-contained folder with HTML, CSS, and JS. The **shared/** folder holds the master theme.css that ALL pages link to — meaning a single edit changes the site-wide color scheme. This separation makes the codebase easy to maintain and extend.

9. Theme System

Centralized branding made easy

One of the system's strongest design decisions is its **single-source-of-truth theme system**. Every page across all four portals (Public, Student, Supervisor, Admin) imports a single `theme.css` file containing every color, shadow, and font in the project.

Active Theme: AWKUM Official

The system currently uses Abdul Wali Khan University Mardan's official brand colors:

Primary `#8B0101`	**Primary Dark** `#5C0000`	**Secondary** `#B82828`	**Gold** `#C9A227`
Gold Light `#DDB94A`	**Background** `#FAF7F7`	**Border** `#E5DCDC`	**Text** `#1A1A1A`

Available Alternative Themes

A separate Theme Palette PDF documents 8 alternative themes that can be applied with a single file edit: Royal Navy, Royal Purple, Ocean Teal, Burgundy & Cream, Slate & Amber, Midnight Indigo, Sunset Maroon, and Charcoal & Gold.

10. Special Features

What makes this system stand out

Picture-in-Picture Video

A floating promotional video plays in the bottom-right corner of every public page. Users can mute/unmute, pause/play, expand to full-screen with a backdrop, drag the player around the screen, scrub through the timeline with a gold seek bar, and close it entirely. Press **Esc** to exit fullscreen, use ← / → arrows to skip 5 seconds.

Live Search Everywhere

Both the Supervisor and Admin dashboards include real-time search functionality. Type a Team ID, Student ID, project title, or student name and results filter instantly with **matched text highlighted in gold**. Empty states show friendly 'no results' messages.

Save-Your-ID Reminders

Whenever an ID is auto-generated (Student ID, Team ID, or Supervisor ID), a prominent **gold-bordered notice** appears next to it telling the user: *'This is your unique ID. Please remember and save it carefully — you will need it for future logins and FYP activities.'* This proactive UX prevents lost IDs.

Hot-Swappable Themes

The entire site — every page, every dashboard — can be re-themed by editing a single CSS file. 9 ready-to-use themes are documented (AWKUM Official, Royal Navy, Royal Purple, Ocean Teal, Burgundy & Cream, Slate & Amber, Midnight Indigo, Sunset Maroon, Charcoal & Gold).

Live Marks Validation

When supervisors enter marks, the inputs **validate in real time** — values exceeding the maximum flash red, empty fields show as 'pending' with dashed borders, and the grand total recalculates as you type. No way to accidentally submit invalid scores.

■ Drag-and-Drop File Uploads

Students can drag files directly onto the upload zones. Supports PDF, DOCX, ZIP, images, and code files (Python, JavaScript, etc.) up to 20MB each. Real-time file size formatting and removal options included.

11. ID Generation System

How unique IDs are created

The system auto-generates three types of IDs, each following a strict format that's easy to recognize and reference.

Type	Example	Format	When Generated
Student ID	`STD-20264731`	`STD-{academic-year}{4-digit-random}`	Generated when a student first registers. Used in all team assignmen
Team ID	`TEAM-20268372`	`TEAM-{academic-year}{4-digit-random}`	Generated when the team leader completes Step 1 of the setup wiza
Supervisor ID	`SUP-20260001`	`SUP-{academic-year}{4-digit-sequential}`	Generated when an admin adds a new supervisor. Sequential within

Important

Once generated, IDs are **permanent** and cannot be modified. They serve as the primary key linking students to teams, teams to projects, and supervisors to teams. Losing track of these IDs makes it harder to look up records — which is why the system shows save-this-ID reminders prominently in the UI.

12. How to Use the System

Quick start guide

For Students

1. Visit the public site and review the **Coursera Requirements** and **FAQs** pages.
2. Once eligible, click **Login** in the top-right and authenticate with student credentials.
3. Complete the **3-step team setup wizard**: project title → team members → supervisor.
4. Save your auto-generated **Student ID** and **Team ID** — you'll need them later.
5. From the **My Project** tab, upload your proposal PDF and wait for supervisor approval.
6. Throughout the semester, attend monthly meetings, submit progress reports, and prepare for vivas.
7. Track your marks anytime in the **Progress & Evaluation** tab.

For Supervisors

1. Receive your **SUP ID** and temporary password from the admin.
2. Log in and immediately note your SUP ID (a notice will appear telling you to save it).
3. Browse your **Dashboard** to see all assigned groups and students at a glance.
4. Use the **search bars** to quickly find a specific team or student.
5. Click any team to open its detail view, then expand each student to enter marks.
6. Save marks frequently — the system validates inputs in real time.

For Admins

1. Log in with admin credentials.
2. Use **Add Supervisor** to register new faculty members. The system generates IDs and passwords automatically.
3. Click any supervisor card on the **Supervisors** view to see their teams and students.
4. Edit any student's marks directly — admin has full edit access.
5. Use the **Teams** view to reassign teams to different supervisors when needed.
6. Monitor the system-wide statistics to identify pending reviews and bottlenecks.

13. Future Enhancements

Roadmap for the next phases

Version 1.0 covers the core FYP workflow. Here are features identified for future iterations:

Short-term (v1.1)

- Backend integration with Node.js + Express + MongoDB for persistent data storage
- Real authentication with JWT tokens and password hashing
- Email notifications for proposal status changes and deadline reminders
- Export marks as PDF/Excel for record-keeping
- In-app messaging between students and supervisors

Medium-term (v1.5)

- Plagiarism integration with Turnitin API for proposal/documentation checks
- Calendar sync (Google Calendar / Outlook) for automatic deadline reminders
- Automated meeting attendance via QR code check-in
- Group chat per team with file sharing
- Comprehensive analytics dashboard for the department head

Long-term (v2.0)

- Multi-department support (Mathematics, Physics, Engineering, etc.)
- Public alumni FYP archive showcasing past projects
- Mobile native apps (Flutter or React Native)
- AI-powered project topic suggestions based on student skill profiles
- Industry partnership module for sponsored FYP projects
- Plagiarism & originality scoring built directly into upload pipeline

14. Conclusion

Closing thoughts

The **FYP Management System** represents a meaningful leap forward in how the AWKUM Department of Computer Science manages its Final Year Projects. By replacing fragmented manual processes with a unified, role-based digital platform, it saves time for everyone involved — students focus on building their projects, supervisors focus on guiding their teams, and administrators get the oversight they need without drowning in paperwork.

Built entirely with vanilla web technologies (HTML, CSS, JavaScript), the system is **lightweight, fast, and easy to extend**. Its modular architecture — with separate folders for each role and a centralized theme system — means adding new features or rebranding the site takes minutes instead of days. The 200-mark evaluation rubric ensures fair, consistent grading across all projects.

Beyond the technical implementation, this project taught the development team valuable lessons in **UX design, state management, responsive layouts, and software architecture**. It also delivered something genuinely useful for our department — a tool that real students, supervisors, and administrators can rely on every day.

Thank You

for reviewing our Final Year Project.

Muhammad Sudais
Full-Stack Developer

Atiya Inayat
Developer

Mehboob Ali
Frontend Developer

Department of Computer Science · Abdul Wali Khan University Mardan
Final Year Project · Academic Year 2025–2026